

Issue 128: A Machine-Certified $4.050498 \times$ Trotter Resource Reduction for a 2D Heisenberg Benchmark

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Abstract

We give an exact-rational, independently checkable certificate for simulating the periodic 12×12 spin-1/2 isotropic Heisenberg model to global operator-norm error at most 10^{-6} for time $T = 1$. The circuit is the same five-copy fourth-order Suzuki product formula, with the same four matching fragments and the same compilation cost model, on both sides of the comparison. A pinned instantiation of the published high-order commutator bound requires 393 steps and 11,791 merged group exponentials. Our certificate accepts 97 steps and rejects 96, requiring 2,911 merged group exponentials. The exact improvement is $11791/2911 = 4.050498110614909\dots$, with a 75.3117% resource reduction. The gain comes from delaying norm inequalities: we compute the formula logarithm, project it to Lie elements, evaluate the concrete lattice representation, combine identical Pauli strings, and certify pairwise-anticommuting groups before closing a finite-step right-generator ledger with a rational tail. The result exceeds the challenge's twofold requirement by a factor of 2.025. An exact degree-five side study is also included, but no 78-step global error certificate is claimed or supplied.

1 Claim boundary and benchmark

The certified benchmark is

$$H = \sum_{\langle i,j \rangle} \frac{X_i X_j + Y_i Y_j + Z_i Z_j}{4}, \quad L = 12, \quad N = L^2 = 144, \quad T = 1, \quad (1)$$

with periodic boundary conditions and target $\|U(T) - \tilde{U}(T)\| \leq 10^{-6}$. Horizontal and vertical bonds are split by parity into four commuting matchings. A symmetric second-order step is composed five times to obtain the standard fourth-order Suzuki formula. After adjacent equal fragments are merged, an r -step simulation contains

$$G(r) = 30r + 1 \quad (2)$$

group exponentials. Each group has $N/2 = 72$ disjoint bond propagators, and we use the same upper bound of three CNOT gates per Heisenberg bond propagator on both sides.

This report makes one principal claim: the supplied exact certificate proves that $r = 97$ is sufficient, while its own error function rejects $r = 96$. The comparison fixes the Hamiltonian, product formula, fragmentation, time, tolerance, and cost model. It therefore measures a proof/certification gain, not a change in physical problem or a favorable redefinition of cost.

The published control is an interval instantiation of the high-order commutator framework of Childs, Su, Tran, Wiebe, and Zhu [1], using the expansion-first local Pauli- ℓ_1 procedure also connected to later

commutator-scaling work [2]. All 31 admissible theorem centers were scanned; center 20 gives the lowest pinned control, 393 steps. This control is stored in the main certificate rather than recomputed from a floating-point constant.

2 Where the improvement comes from

Let the four anti-Hermitian matching generators be $A_1, \dots, A_4$, and let $S_4(t)$ denote one merged fourth-order step. A conventional bound expands partial Hamiltonian sums into many nested commutators and applies triangle inequalities early. That destroys cancellations before the concrete Heisenberg algebra is visible. Our method reverses this order:

1. compute the homogeneous free-associative-word logarithm of the complete 31-stage formula;
2. use the Dynkin–Specht–Wever map to project the degree-five and degree-seven pieces into their Lie elements;
3. evaluate those Lie elements in the four-matching Heisenberg representation on the infinite translation cell;
4. canonicalize translations and combine identical Pauli strings before any final norm inequality;
5. partition the leading right-generator terms into exactly verified pairwise-anticommuting groups; and
6. convert the logarithm to the right generator, retain degrees four through seven explicitly, and bound the remaining degrees by a rational geometric locality tail.

If the Pauli strings in $K_g = \sum_j c_j P_j$ anticommute pairwise, then $K_g^2 = (\sum_j c_j^2)I$, and hence

$$\|K_g\| = \sqrt{\sum_j |c_j|^2}. \quad (3)$$

We use an outward rational upper bound for each square root. The greedy grouping algorithm is only a discovery tool: the sidecar records every term, every group membership, and every group bound. The verifier independently checks exact coverage and every symplectic anticommutation relation. Thus a different grouping heuristic can change efficiency, but it cannot weaken the soundness of an accepted certificate.

The submitted leading-defect sidecar contains 75,324 canonical Pauli terms in 7,576 groups, with maximum group size 10. The direct termwise cell bound 20.160968407335066 is reduced to the certified grouped bound 6.472926505087888, an improvement of 3.114660484942633 at the leading local norming step.

3 A Heisenberg-specific local lemma

The higher-degree ledger also benefits from an exact local observation. For one normalized bond

$$h = \frac{XX + YY + ZZ}{4}, \quad (4)$$

every phase-free two-qubit Pauli string anticommutes with either zero or exactly two of XX , YY , and ZZ . Direct enumeration of the 16 local Pauli cases therefore gives

$$\|[h, P]\|_{\ell_1, \text{Pauli}} \leq \|P\|_{\ell_1, \text{Pauli}}, \quad (5)$$

so the bond commutator growth constant is 1 rather than the generic triangle-inequality value $3/2$. This is a small lemma with a large cumulative effect because it enters repeated support-growth estimates. The verifier checks the complete local table exactly; it does not assume the lemma as an external analytic input.

This separation of discovery and trust is important. Symbolic search finds a compact representation and a strong grouping, while the trusted path uses integers, rational numbers, outward rational intervals, and exact symplectic Pauli multiplication. Floating point is used only to rank candidates, print human-readable decimals, and perform a small dense sanity check.

4 Finite-step error certificate

Write the one-step logarithm as

$$\log S_4(t) = tA + t^5B + t^7C + O(t^9), \quad (6)$$

where $A = \sum_{j=1}^4 A_j$. The exact right logarithmic generator has the low-degree expansion

$$S'_4(t)S_4(t)^{-1} - A = 5t^4B + 2t^5[A, B] \quad (7)$$

$$+ t^6 \left(7C + \frac{2}{3}[A, [A, B]] \right) \quad (8)$$

$$+ t^7 \left(3[A, C] + \frac{1}{6}[A, [A, [A, B]]] \right) + R_{\geq 8}(t). \quad (9)$$

The D4 term is evaluated with the exact anticommuting partition. Degrees D5 through D7 are bounded by the certified concrete coefficients and the local lemma. Support growth, conjugation Taylor factorials, and the Duhamel integration factor make $R_{\geq 8}$ an explicit rational geometric series. Nothing beyond degree seven is silently discarded.

For $r = 97$, the exact rational contributions have the following outward decimal renderings.

Table 1: Certified global operator-norm error ledger at $r = 97$.

Right-generator contribution	Global upper bound
Degree 4	$5.264367936822258 \times 10^{-7}$
Degree 5	$1.383487536744612 \times 10^{-7}$
Degree 6	$1.579057000594835 \times 10^{-7}$
Degree 7	$1.894657080691470 \times 10^{-8}$
Degree 8 and above	$1.542560312083473 \times 10^{-7}$
Total	$9.958938494314325 \times 10^{-7}$

The underlying total is a rational number and is strictly below 10^{-6} . At $r = 96$, the outward decimal is $1.050565873970784 \times 10^{-6}$, strictly above the target. The step boundary is checked by cross-multiplying integers, not by comparing rounded decimals. The printed decimal convention rounds positive certified upper bounds upward at 16 significant digits, so display conversion never understates the bound.

5 Resource result

Because both rows use the same 31-stage formula and compilation model, the resource comparison follows directly from Eq. (2).

Table 2: Pinned rigorous control and certified result.

Resource	Published control	This work	Reduction
Trotter steps	393	97	75.32%
Merged group exponentials	11,791	2,911	75.31%
Bond propagators	848,952	209,592	75.31%
CNOT upper bound	2,546,856	628,776	75.31%

The primary exact ratio is

$$\frac{G(393)}{G(97)} = \frac{11791}{2911} = 4.050498110614909\dots \quad (10)$$

The challenge asks for at least a twofold resource improvement. The achieved ratio is 2.025249 times that requirement—equivalently, 202.5% of the stated minimum ratio, or 102.5% beyond it. We call this a fourfold certified result, not a fivefold result.

6 Independent checks and negative results

The certificate is designed to be reviewable without trusting the search code. The fast verifier checks the main certificate digest bindings, all 75,324 D4 terms, exact coverage by 7,576 groups, every pairwise anticommutation relation, each rational square-root upper bound, the integer step boundary, and all resource arithmetic. A deep mode regenerates the published control, the canonical leading-defect coefficients, the degree-seven majorant, and the finite tail from the formula.

A degenerate 2×2 periodic-algebra cross-check gives actual dense spectral-norm error $9.128277711816321 \times 10^{-10}$, below its outward certificate $2.766666666666667 \times 10^{-8}$. This is not used to infer the 12×12 result; it catches representation, stage-ordering, and normalization errors on a tractable instance.

We also record routes that did not justify a stronger claim:

- The degree-three Lie representation has exact rank 20, equal to the free-Lie dimension $(4^3 - 4)/3 = 20$. The four matching fragments thus offer no free model-specific reduction of fourth-order conditions.
- Standard recursive Suzuki orders 2, 4, 6, and 8 compile to 269,677, 36,361, 83,101, and 377,251 groups, respectively, under the audited generic locality-tail bound. Raising recursive order alone is worse.
- A three-site $SU(2)$ cluster decomposition improves local block counts but not the conservative compiled CNOT bound after including the three-site propagator cost.
- Empirically optimized formulas were not substituted for a deterministic worst-case certificate. Their sampled constants cannot replace an interval-certified comparison on this benchmark.

7 Fivefold feasibility boundary

Fivefold arithmetic would require $G(r) \leq \lfloor 11791/5 \rfloor$, hence $r \leq 78$. Conditional on an accepted 78-step error proof, $G(78) = 2341$ and $11791/2341 = 5.036736437419906\dots$. That arithmetic is not an error certificate.

As a concrete follow-up, we generated and independently verified an exact degree-five sidecar with 605,832 canonical Pauli terms, 123,106 same-support pairwise-anticommuting groups, and site-density upper bound 11.23706750025696. This closes one useful algebraic subproblem and validates the streamed cubic-field machinery. It does not close the global ledger at $r = 78$: the translation-coupled D4 budget and explicit D8-plus delayed-tail certificate remain hard gates. **No 78-step global error certificate is claimed or supplied.**

The audit therefore gives a precise next research decision. Rebuild the complete exact ledger at $r = 78$ only after D4 and D8-plus gates are closed; if the sum is at most 10^{-6} , fivefold can be claimed. For tenfold, the current fourth-order ledger is off by hundreds even with the D4 contribution removed, so an effective order lift—for example a certified processor or a shorter sixth-order kernel—is required.

8 Reproduction and delivered evidence

From the `issue128` directory, a reviewer can run

```
python -m pip install -e '[test]'
pytest -q
PYTHONPATH=src python scripts/verify.py \
    certificates/issue128-certificate.json
PYTHONPATH=src python scripts/build_d5_certificate.py --verify-only
PYTHONPATH=src python scripts/package_delivery.py --check
shasum -a 256 -c artifacts/SHA256SUMS
```

The delivery contains: (i) the main exact-rational certificate; (ii) D4 and D5 sidecars bound by SHA-256; (iii) machine-readable JSON and plain-text result summaries; (iv) a captured verification transcript with exit codes; (v) this report and its source; (vi) regression tests; and (vii) one checksum manifest binding all reviewer-facing evidence. The repository and pull request are the transport mechanism; the numerical claim is anchored in the exact artifacts and executable verification path.

The three large proof inputs have fixed identities:

Artifact	SHA-256
Main certificate	0a09623ce3b292a3637065c870fb3153bbdcddce30aef968565c4db3ddfc7201
D4 sidecar	a397414bb0229fb1ebdb38798aa781fb89dbb9d5cdbed94c7cd2e9120da62718
D5 sidecar	c5e8968a93b4497b41fe42c0e364324388272e183e1fcd20a536bc988f5361dd

The committed `artifacts/SHA256SUMS` file additionally binds the final PDF, report source, summaries, transcript, cross-check, and sidecars as one reviewer-facing package.

9 Conclusion

Issue 128’s basic requirement is fully met for the fixed benchmark. The certified resource count drops from 11,791 to 2,911 merged group exponentials, an exact 4.050498-fold improvement, without changing the circuit family or cost model. The central algorithmic contribution is not merely a faster implementation: it is a tighter proof architecture that preserves concrete Lie/Pauli cancellation until exact local algebra and anticommutation can be used, then closes the remaining finite-step error with rational arithmetic. The supplied data, transcript, tests, and PDF are mutually hash-bound and independently reproducible. Fivefold remains a clearly delimited follow-up, not part of the certified result.

References

- [1] Andrew M. Childs, Yuan Su, Minh C. Tran, Nathan Wiebe, and Shuchen Zhu. A theory of trotter error. *Physical Review X*, 11(1):011020, 2021.
- [2] Ansgar Schubert and Christian B. Mendl. Trotter error with commutator scaling for the Fermi–Hubbard model. *arXiv preprint arXiv:2306.10603*, 2023.